

* SIMPLE DISTILLATION *

1. INTRODUCTION

Simple distillation is the simplest method of distillation. It is used to separate a liquid from non-volatile impurities or to separate two miscible liquids having large difference in boiling points ($\Delta T_b > 25-30^\circ\text{C}$).

2. PRINCIPLE

- The feed is heated in still.
- The more volatile component vapourizes first.
- Vapours rise and pass through the condenser.
- In the condenser, vapours are cooled and condensed.
- Condensed liquid (distillate) is collected in receiver.
- Less volatile component remains in still as residue.

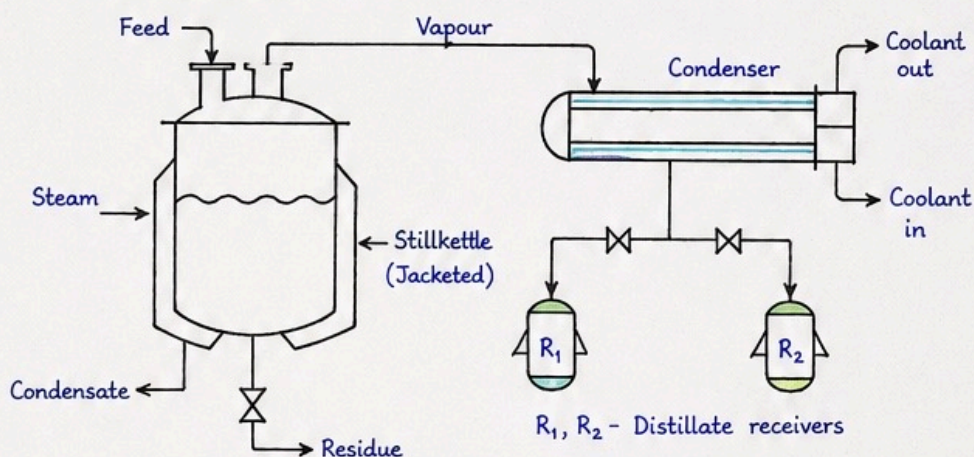
4. WORKING

1. Feed is charged into still and steam is passed through jacket.
2. Mixture is heated. More volatile component vapourizes.
3. Vapours pass to condenser.
4. Cooling water enters at bottom and leaves at top of condenser.
5. Vapours condense and liquid collects in receivers.
6. Less volatile component remains in still as residue.

6. ADVANTAGES

- 😊 Simple construction and easy operation.
- 😊 Low initial cost.
- 😊 Suitable for large difference in boiling points.
- 😊 Easy to clean and maintain.
- 😊 Can handle high feed rates.

3. NEAT, LABELLED DIAGRAM



Feed - Mixture to be separated
 Steam - For heating the still (through jacket)
 Residue - Bottom product (less volatile)
 Distillate - Top product (more volatile)
 Coolant - Cooling water

5. APPLICATIONS

- Purification of liquids.
- Separation of liquids with large difference in boiling points.
- Recovery of solvents.
- Desalination of water.

7. LIMITATIONS

- ☹ Not suitable for close boiling point liquids.
- ☹ Low separation efficiency.
- ☹ More energy consumption.
- ☹ Not used for azeotropic mixtures.

* FRACTIONAL DISTILLATION *

Fractional distillation is a process used to separate a liquid mixture into its components having close boiling points by using a fractionating column.

① PRINCIPLE :-

It is based on difference in volatility of components. More volatile component rises in vapour phase and less volatile component remains in liquid phase. Repeated contact of vapour and liquid on trays/packing gives better separation.

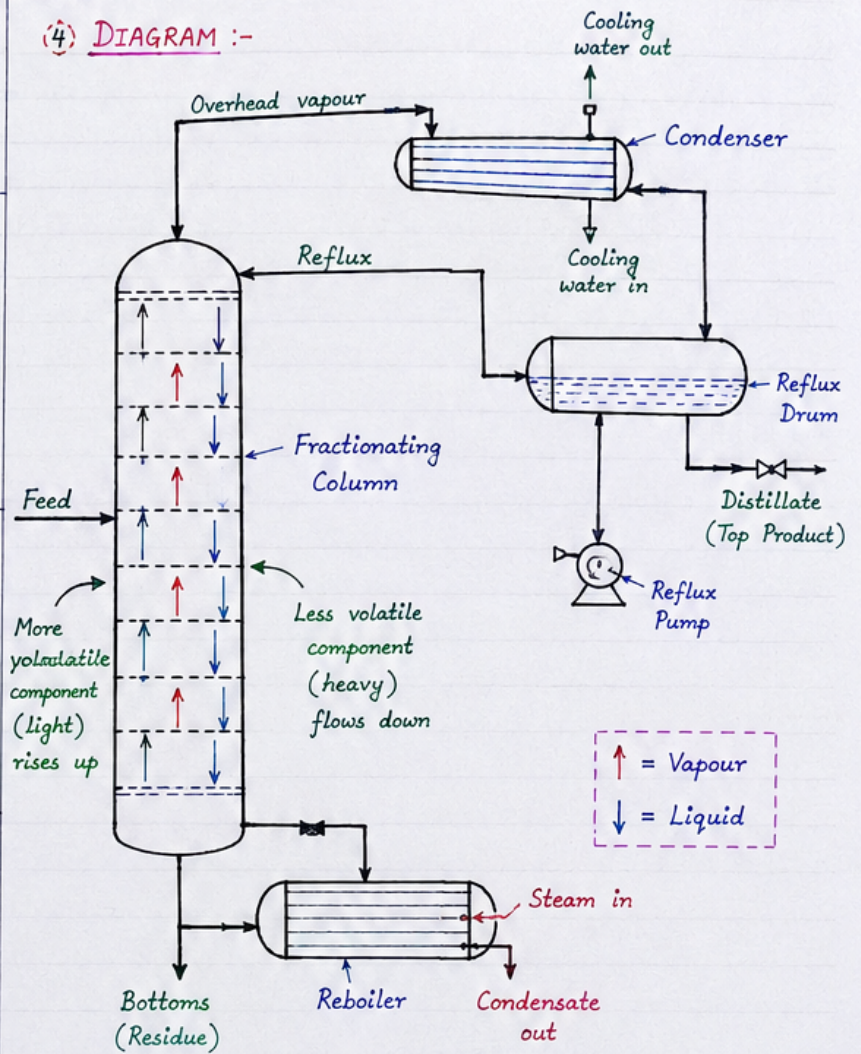
② USES :-

- Separation of liquids having close boiling points.
- Petroleum refining (crude oil)
- Solvent recovery
- Alcohol purification
- Chemical industries

③ MAIN EQUIPMENT :-

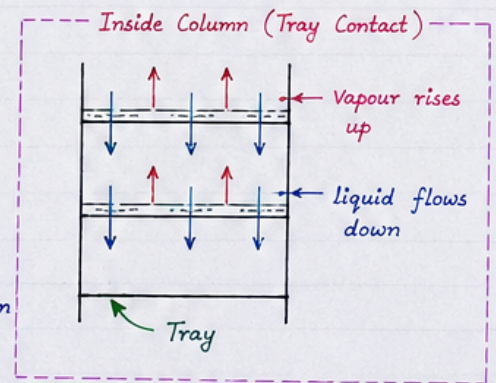
- ① Fractionating Column
- ② Reboiler
- ③ Condenser
- ④ Reflux Drum
- ⑤ Reflux Pump
- ⑥ Trays / Packing
- ⑦ Feed Inlet
- ⑧ Bottoms Outlet
- ⑨ Distillate Outlet

④ DIAGRAM :-



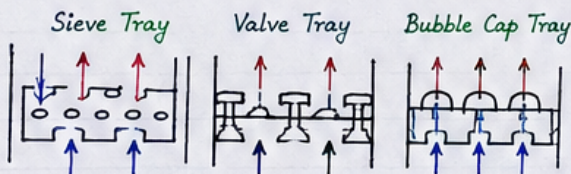
⑤ WORKING :-

- ① Feed is introduced into the column.
- ② Reboiler heats the bottom liquid and vapour is generated.
- ③ Vapour rises upward through the column.
- ④ Liquid from upper trays flows downward.
- ⑤ On each tray/packing, vapour and liquid contact each other.
- ⑥ More volatile component goes to top, less volatile goes to bottom.
- ⑦ Top vapour is condensed in condenser.
- ⑧ Condensed liquid collected in reflux drum.
- ⑨ A part of it is returned to column as reflux and rest is withdrawn as distillate (top product).
- ⑩ Bottom liquid is withdrawn as bottoms (residue).



⑥ TRAYS TYPES :-

- ① Sieve Tray
- ② Valve Tray
- ③ Bubble Cap Tray



⑧ ADVANTAGES :-

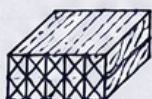
- Can separate liquids having close boiling points.
- High purity of products.
- Continuous operation possible.
- Better efficiency.

⑦ PACKING TYPES :-

- ① Random Packing - Raschig Ring, Pall Ring, Saddle



- ② Structured Packing - (e.g. Structured sheet, Mellapak)



⑨ LIMITATIONS :-

- High initial cost.
- More energy requirement.
- Operation is complex.
- Flooding, weeping, entrainment etc. may occur.

Key Point :-

- More volatile component → Top (Distillate)
- Less volatile component → Bottoms (Residue)

FLASH DISTILLATION

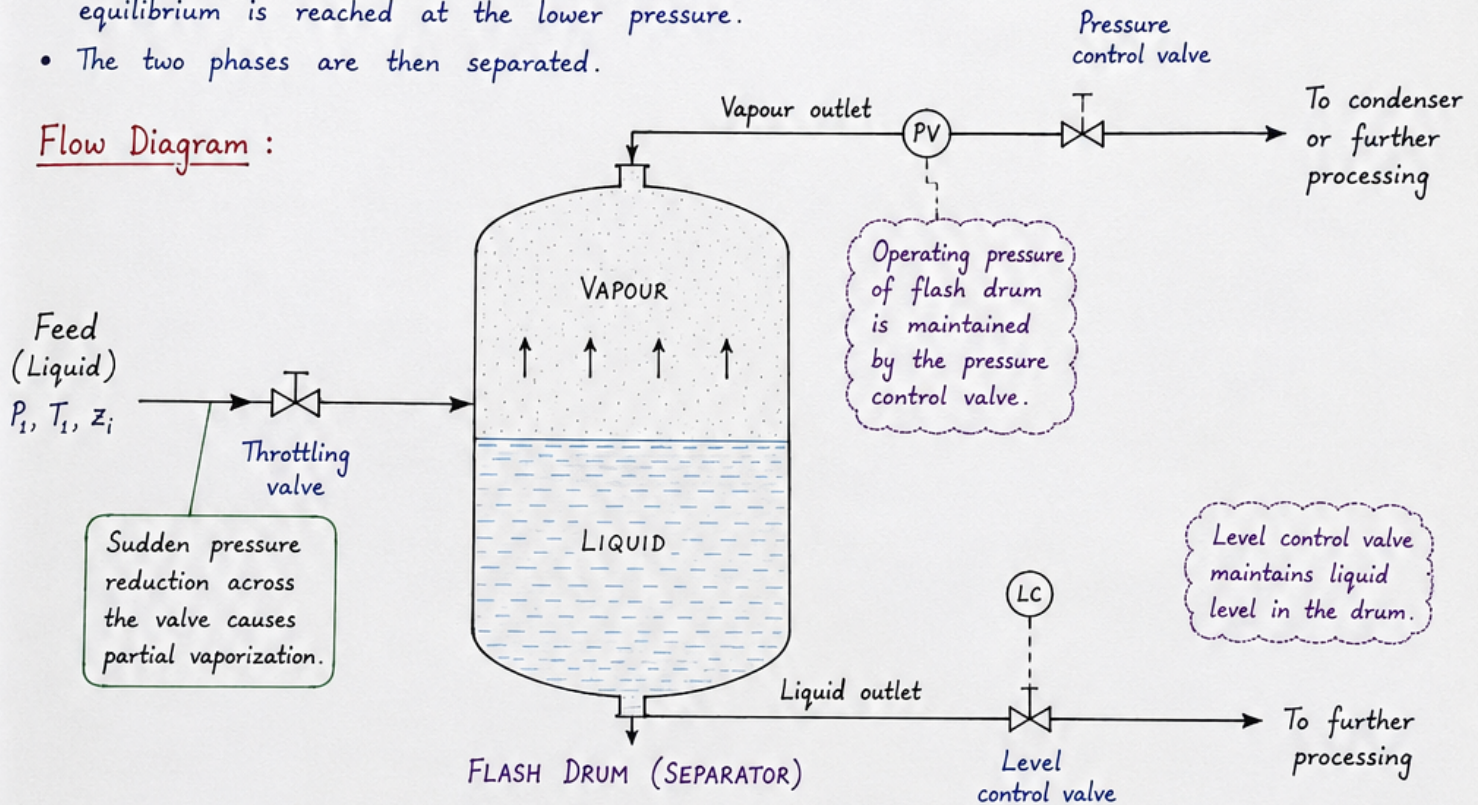
Definition :

Flash distillation (or flash vaporization) is a single-stage equilibrium separation process in which a feed is partially vaporized by a sudden reduction in pressure (at nearly constant temperature), producing a vapour and a liquid stream in equilibrium.

Principle :

- When the pressure of a liquid mixture is reduced suddenly (throttling) its boiling point decreases.
- Part of the liquid "flashes" into vapour until vapour-liquid equilibrium is reached at the lower pressure.
- The two phases are then separated.

Flow Diagram :



Key Points :

- It is a single stage, equilibrium unit operation.
- No heat is added. The energy required for vaporization is obtained by the sensible heat of the feed.
- The operation is adiabatic (no heat exchange).
- The vapour and liquid leaving the drum are in thermodynamic equilibrium.

Applications / Uses :

- Separation of light ends from heavy hydrocarbon streams in petroleum refineries.
- Removal of dissolved gases from liquids.
- Pre-treatment of feeds before further separation (distillation, absorption, etc.).
- Used in evaporation, degassing and stabilizer units.

Material Balance :

Let,

F = Feed flow rate (kmol/h)

V = Vapour (flash) flow rate (kmol/h)

L = Liquid flow rate (kmol/h)

z_i = Mole fraction of component i in feed

y_i = Mole fraction of component i in vapour

x_i = Mole fraction of component i in liquid

Total Balance : $F = V + L$

Component Balance : $Fz_i = Vy_i + Lx_i$

Flash (Vapour) Fraction :

$$\frac{V}{F} = \frac{h_F - h_L}{h_V - h_L}$$

Where,

h_F = Enthalpy of feed

h_V = Enthalpy of vapour

h_L = Enthalpy of liquid

(All at flash drum pressure)

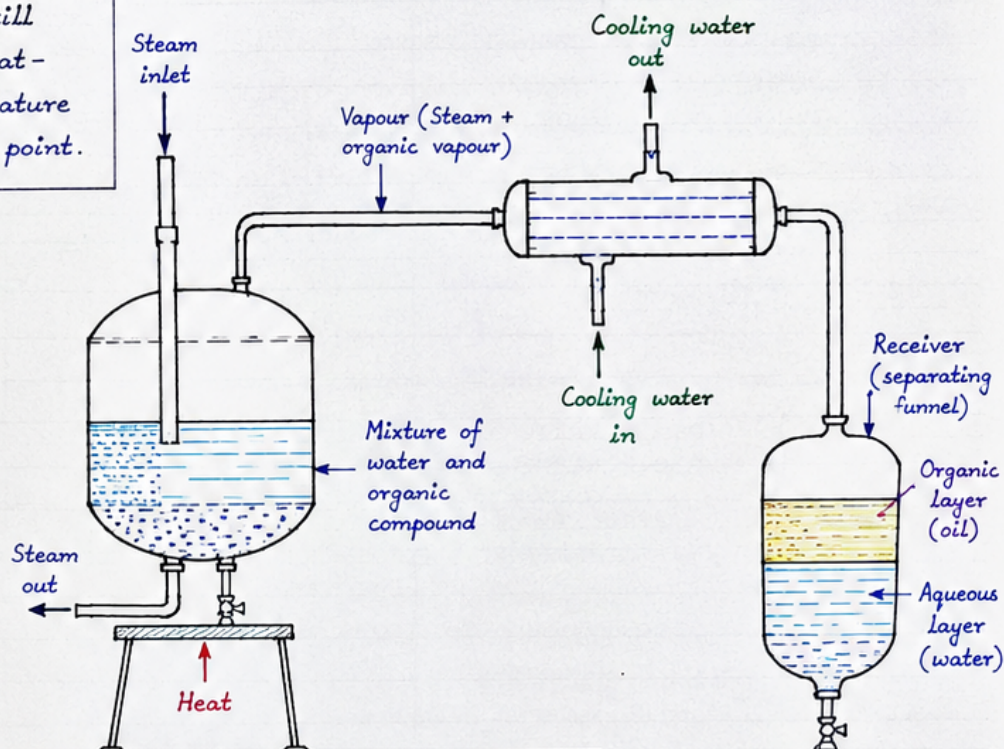
Important Notes :

- ① Flash distillation differs from simple distillation as it has no trays or packing.
- ② The separation depends on pressure drop and feed condition.
- ③ Lower operating pressure $\Rightarrow$ more vapour formed.
- ④ The two outlet streams are in equilibrium at the drum pressure.

* STEAM DISTILLATION *

Steam distillation is a separation process in which steam is used to distill volatile, water-immiscible and heat-sensitive compounds at a temperature lower than their normal boiling point.

(1) DIAGRAM :-



(2) PRINCIPLE :-

- Steam and the organic compound are immiscible.
- The total vapour pressure of the mixture is equal to the external pressure at a temperature lower than the normal boiling point of the organic compound.
- Hence the mixture boils and volatile compounds distil over with steam.

(3) WORKING :-

- ① Steam is generated in a boiler and introduced into the distillation vessel.
- ② Steam passes through the material (mixed with water or supported on a tray).
- ③ Volatile, water-immiscible compounds vaporize along with steam.
- ④ The vapour mixture (steam + organic vapour) passes through the condenser.
- ⑤ In the condenser, vapour is cooled and condensed to liquid.
- ⑥ The distillate is collected in the receiver. Since the organic compound and water are immiscible, two separate layers are formed.
- ⑦ Organic layer (oil) is lighter or heavier than water depending on the compound and is separated.

8. KEY NOTES :-

- More volatile compound co-distills with steam.
- Lower boiling point due to increase in total vapour pressure.
- Immiscibility between water and organic compound is important.

4. TYPES :-

- ① Direct Steam Distillation: Steam is passed directly into the material.
- ② Steam and Water Distillation: Material is immersed in water and steam is passed.
- ③ Water Distillation: Material is boiled directly with water.

5. APPLICATIONS :-

- Extraction of essential oils (eucalyptus, peppermint, rose oil, etc.)
- Heat sensitive materials
- Purification of volatile organic compounds
- Pharmaceutical and chemical industries
- Extraction of flavours and fragrances

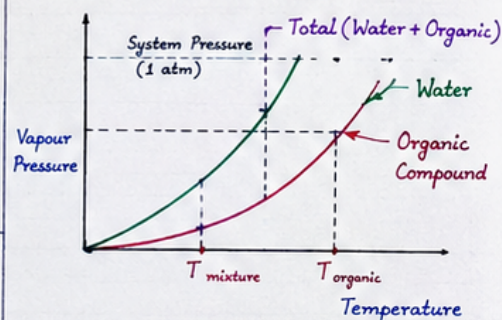
6. ADVANTAGES :-

- ✓ Distillation at lower temperature.
- ✓ Suitable for heat-sensitive materials.
- ✓ Prevents decomposition of compounds.
- ✓ Simple and easy operation.
- ✓ Good for water-immiscible volatile compounds.

7. LIMITATIONS :-

- ✗ Compound must be volatile with steam.
- ✗ Not suitable for water-soluble compounds.
- ✗ Large quantity of steam is required.
- ✗ Not suitable for very high boiling compounds.

9. TEMPERATURE CONCEPT :-



$$T_{\text{mixture}} < T_{\text{organic}}$$

Hence boiling occurs at lower temperature.

VACUUM DISTILLATION

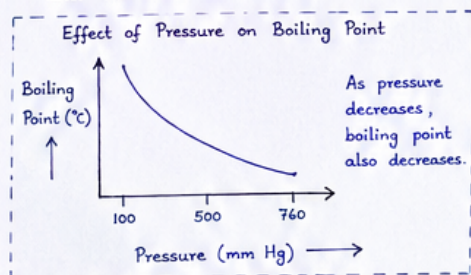
Pressure ↓
Boiling Point ↓
Temperature ↓

1. DEFINITION :-

Vacuum distillation is a distillation process carried out at a pressure lower than atmospheric pressure. Due to reduced pressure, the boiling point of the liquid decreases and high boiling / heat sensitive liquids can be distilled at lower temperature.

2. PRINCIPLE :-

A liquid boils when its vapour pressure becomes equal to the surrounding pressure. If external pressure is reduced by vacuum pump, the boiling point of the liquid decreases. The vapour formed is condensed and distillate is collected.



3. WHY VACUUM DISTILLATION IS USED ?

- * For high boiling point liquids.
- * For heat sensitive / thermally unstable materials.
- * To prevent decomposition or unwanted reactions.
- * To distil materials at lower temperature.
- * To improve product quality.

4. MAIN COMPONENTS :-

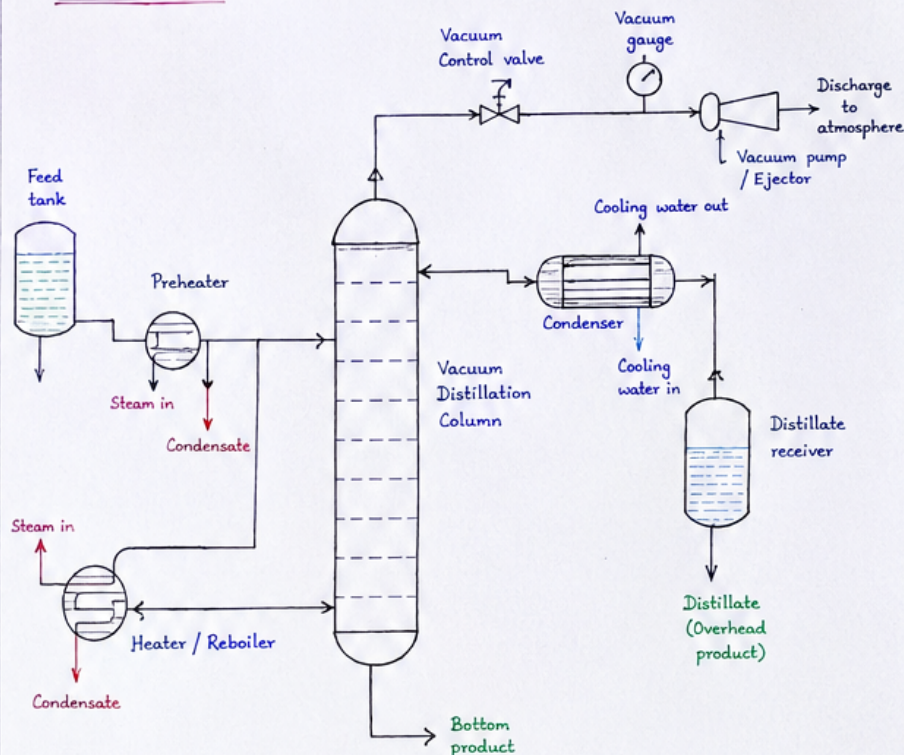
1. Feed tank
2. Preheater (optional)
3. Heater / Reboiler
4. Vacuum distillation column
5. Condenser
6. Distillate receiver
7. Vacuum pump / Ejector
8. Vacuum gauge
9. Pressure control system
10. Cooling water system
11. Vacuum breaker
12. Bottom product outlet

All components must be properly sealed to maintain vacuum.

5. WORKING :-

1. Feed is taken from feed tank.
2. It may be preheated in preheater (if required).
3. Then it enters the heater / reboiler where heat is supplied.
4. Vacuum pump removes non-condensable gases and reduces the pressure in the system.
5. Due to low pressure, liquid boils at lower temperature and vapours are generated.
6. Vapours rise through the column.
7. In the column, vapour and liquid contact each other on trays or packing → separation takes place.
8. Overhead vapours go to the condenser and get condensed.
9. Condensed liquid is collected in distillate receiver.
10. Heavier / less volatile liquid leaves from the bottom as bottom product.

6. FLOW DIAGRAM :-



7. IMPORTANT RELATIONSHIP :-

$$\text{Pressure} \downarrow \rightarrow \text{Boiling point} \downarrow$$

$$P_1 < P_2 \rightarrow T_{b1} < T_{b2}$$

Lower the pressure, lower the boiling temperature of the liquid.

8. COMPARISON :-

Parameter	Atmospheric Distillation	Vacuum Distillation
Pressure	Near atmospheric	Below atmospheric
Boiling temperature	High	Low
Suitable for	Thermally stable materials	Heat sensitive / high boiling materials
Temperature difference	High	Low
Equipment	Simple	Requires vacuum system
Risk of thermal degradation	More	Less

9. APPLICATIONS :-

- Petroleum industry (vacuum gas oil, lube base oils, heavy fractions).
- Pharmaceutical industry (heat sensitive compounds).
- Chemical industry (high boiling chemicals).
- Essential oils and fragrances.
- Solvent recovery.
- Edible oil purification.

10. ADVANTAGES :-

- * Lower boiling temperature.
- * Prevents thermal decomposition.
- * Suitable for heat sensitive materials.
- * Better product quality.
- * Helps in distillation of high boiling liquids.

11. DISADVANTAGES :-

- * Requires vacuum system.
- * Leakage of air disturbs operation.
- * Vacuum equipment is costly.
- * Maintenance of vacuum pump is required.
- * Condensation under vacuum is more difficult.
- * Foaming and entrainment may occur.

12. KEY TERMS :-

Vacuum : Pressure lower than atmospheric pressure.

Absolute pressure : Pressure measured from perfect vacuum.

Reflux : Condensed liquid returned to column.

Distillate : Overhead product after condensation.

Bottom product : Less volatile liquid leaving from bottom.

13. COLUMN CONSIDERATIONS IN VACUUM DISTILLATION :-

- * Vapour volume is high at low pressure → column diameter is usually large.
- * Pressure drop in column must be low.
- * Packing is often used (Raschig rings, Pall rings, structured packing) for low pressure drop and good mass transfer.
- * Trays should provide good contact with minimum pressure drop.



14. VACUUM SYSTEM :-

Vacuum pump / Ejector removes gases and maintains required low pressure.

Types :-

1. Liquid ring vacuum pump
2. Steam ejector
3. Dry vacuum pump
4. Mechanical pump

15. IMPORTANT POINTS (FOR EXAM) :-

1. Pressure ↓ → Boiling point ↓
2. Main purpose is to reduce operating temperature.
3. Useful for high boiling / heat sensitive liquids.
4. Low pressure → vapour volume ↑.
5. Excessive pressure drop increases boiling temperature.
6. Vacuum pump maintains low pressure.
7. Equipment must be well sealed.

In short : Vacuum distillation is a separation process used when high temperature distillation is not possible. It is based on the principle that lowering the pressure lowers the boiling point. *

* AZEOTROPIC DISTILLATION *

Azeotropic distillation is a special distillation used to separate liquid mixtures which form azeotropes and cannot be completely separated by ordinary fractional distillation.

① AZEOTROPE :-

- An azeotrope is a constant boiling mixture.
- At azeotropic composition, the vapour composition is identical to the liquid composition.
- Hence, no change in composition occurs in the distillate and residue beyond this point.
- Types : (i) Minimum boiling azeotrope
(ii) Maximum boiling azeotrope

② PRINCIPLE :-

An additional substance called entrainer is added to the azeotropic mixture. It changes the relative volatility of the components and destroys or shifts the azeotropic point so that the components can be separated.

③ ENTRAINER :-

- A high boiling compound added in small quantity.
- It should be more volatile than one component and less volatile than the other.
- It should form minimum boiling azeotrope with one of the components.
- It should be easily recovered and recycled.
- It should be chemically stable and inexpensive.

④ WORKING (with entrainer) :-

- Feed (azeotropic mixture) and entrainer are fed to the distillation column.
- In the reboiler, mixture is partially vaporised.
- Due to presence of entrainer, relative volatility of key components changes.
- Vapour rises in the column and liquid flows down.
- Overhead vapour is condensed in condenser.
- Condensed liquid goes to decanter where two liquid phases are formed.
- One phase (organic/entrainer rich) is returned to column as reflux.
- Other phase (product rich) is withdrawn as distillate.
- Bottoms may be water rich or heavy component depending on the system.

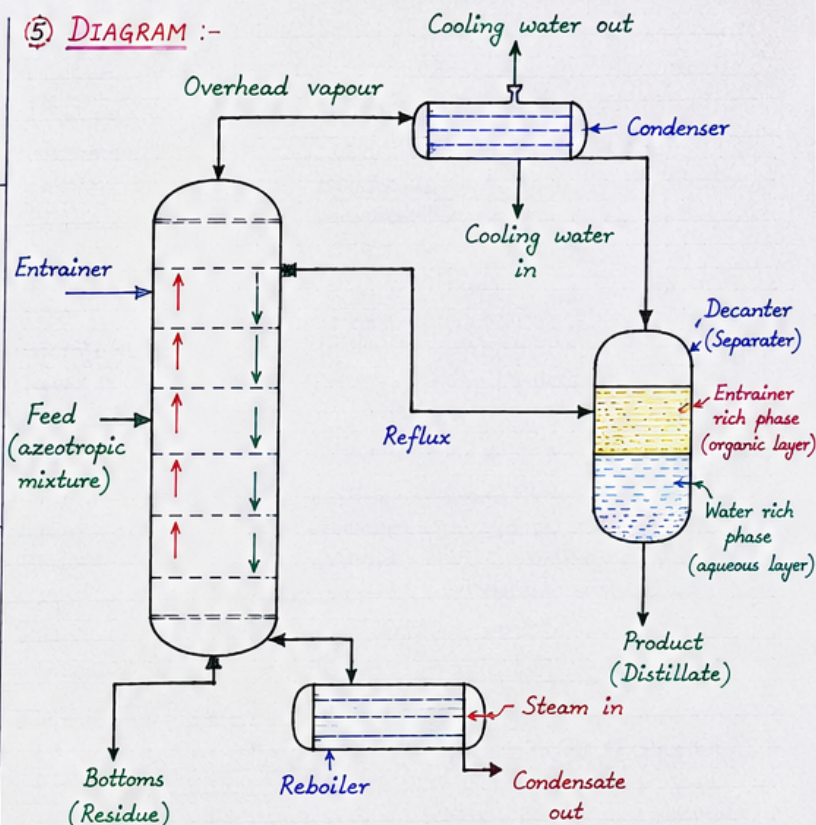
⑧ EXAMPLES :-

- Ethanol - Water system (with benzene as entrainer)
- Isopropanol - Water system (with cyclohexane)
- Acetic acid - Water system
- Nitrobenzene - Water system

⑨ ADVANTAGES :-

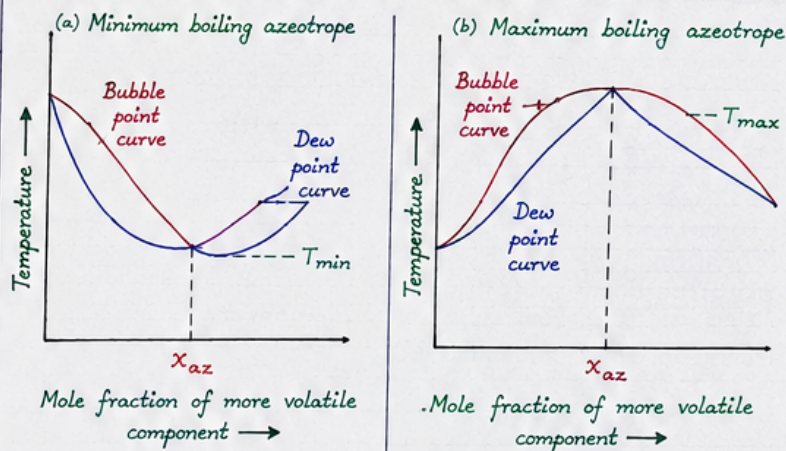
- Separation of azeotropic mixtures is possible.
- High purity of products can be obtained.
- Effective for difficult separations.
- Useful in solvent dehydration and purification.

⑤ DIAGRAM :-



Note :- Two liquid phases form in decanter because entrainer and water are immiscible.

⑥ VAPOUR-LIQUID EQUILIBRIUM (T-x-y DIAGRAM)



⑦ AZEOTROPIC vs ORDINARY DISTILLATION

Basis	Azeotropic Distillation	Ordinary Distillation
Mixture	Forms azeotrope	Does not form azeotrope
Separation	Possible with entrainer	Complete based on volatility
Entrainer	Required	Not required
Relative volatility	Modified by entrainer	Natural volatility
Equipment	Column + Decanter	Column + Condenser
Product	High purity	Depends on system
Complexity	More complex	Relatively simple

⑩ LIMITATIONS :-

- Extra entrainer is required.
- Additional equipment (decanter, entrainer recovery) increases cost.
- Energy consumption is more.
- Entrainer must be recovered and recycled.
- Not suitable if entrainer forms azeotrope difficult to break.

Key Point :- Azeotropic distillation works by changing VLE behaviour of the system using an entrainer so that azeotropic mixture can be separated.

EXTRACTIVE DISTILLATION

Used to separate close boiling mixtures or azeotropic mixtures.

1. DEFINITION :-

Extractive distillation is a separation process in which a high-boiling solvent (entraining solvent) is added to the feed mixture to change the relative volatility of the components and thus achieve separation by distillation.

2. PRINCIPLE :-

The entrainer (solvent) interacts preferentially with one of the components in the feed. This changes the relative volatility between the components, i.e., increases α (relative volatility). As a result, the components can be separated in the distillation column.

Relative Volatility (α)

$$\alpha_{AB} = \left(\frac{y_A/x_A}{y_B/x_B} \right)$$

In extractive distillation $\rightarrow \alpha$ increases from near 1 (difficult separation) to much greater than 1 (easy separation).

3. WHY EXTRACTIVE DISTILLATION IS USED ?

- * For azeotropic mixtures (when azeotrope forms at given pressure).
- * For close boiling mixtures (small α).
- * To improve separation without changing column pressure drastically.
- * To obtain high purity products.

4. TYPES (BASED ON ENTRAINER VOLATILITY)

(a) High-Boiling Entrainer

- * Less volatile than both components.
- * Mostly remains in bottom.
- * Needs solvent recovery column.

(b) Low-Boiling Entrainer

- * More volatile than both components.
- * Leaves with overhead vapour.
- * Needs solvent recovery from distillate.

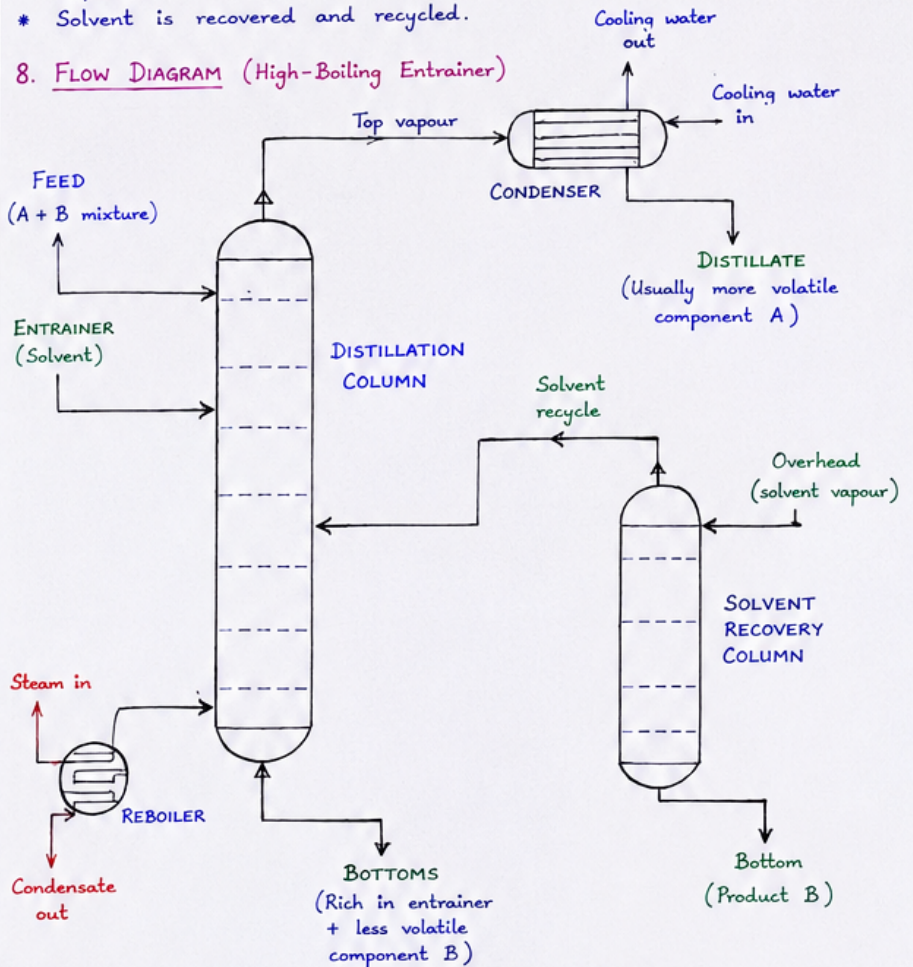
5. PROPERTIES OF A GOOD ENTRAINER :-

1. Selective toward one of the components.
2. High boiling point (usually non-volatile).
3. Chemically stable and inert.
4. Low toxicity and non-corrosive.
5. Easy to separate from products.
6. Affordable and readily available.

7. KEY POINTS :-

- * Entrainer does not appear in the distillate (for high-boiling entrainer).
- * Entrainer changes VLE of the system.
- * More stages or higher reflux may be required.
- * Solvent is recovered and recycled.

8. FLOW DIAGRAM (High-Boiling Entrainer)



9. ADVANTAGES :-

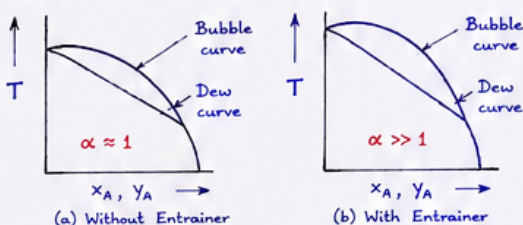
- * Can break azeotropes.
- * Improves relative volatility greatly.
- * High purity products can be obtained.
- * Applicable to a wide range of systems.

10. DISADVANTAGES :-

- * Requires additional solvent.
- * More energy due to solvent recovery.
- * More complex (extra column).
- * Cost of solvent and recovery.

6. EFFECT OF ENTRAINER ON RELATIVE VOLATILITY

Without entrainer $\rightarrow \alpha \approx 1$ (difficult separation)
With entrainer $\rightarrow \alpha \gg 1$ (easy separation)



Entrainer lowers the volatility of the more volatile component or increases volatility of the less volatile component, thus increasing α .

11. AZEOTROPIC VS EXTRACTIVE DISTILLATION

Point	Azeotropic Distillation	Extractive Distillation
Solvent used	Entrainer (may form azeotrope)	Entraining solvent (does not form azeotrope)
Distillate	Entrainer may appear in distillate	Entrainer does not appear in distillate (high boiling)
Overhead	Often needs decanter (if heterogeneous)	Usually homogeneous (no decanter)
Relative volatility	Changes by forming new azeotrope	Changes by altering VLE
Recovery	Decanter + recycle	Solvent recovery column
Use	For azeotropic mixtures	For close boiling or azeotropic mixtures

12. EXAMPLES :-

1. Benzene - Cyclohexane (use Sulfolane)
2. Ethanol - Water (use Glycerol, EG)
3. Acetone - Chloroform (use DMSO)
4. Acetic acid - Water (use Glycols)
5. Many petrochemical separations.

13. COMMON ENTRAINERS :-

1. Glycols (EG, DEG, TEG)
2. Glycerol
3. Sulfolane
4. Dimethyl sulfoxide (DMSO)
5. N-Methyl pyrrolidone (NMP)
6. Acetamide

14. IMPORTANT NOTES :-

- * Entrainer should have very low volatility.
- * Recovery of entrainer is essential for economy.
- * Increasing solvent/feed ratio increases separation but increases cost.
- * Selectivity of entrainer is the key.

In short :- Extractive distillation uses a high boiling solvent to change the relative volatility between components and enables separation of difficult mixtures.

TROUBLESHOOTING & ROOT CAUSE FOR EACH DISTILLATION TYPE

1 SIMPLE DISTILLATION

PROBLEM / SYMPTOM	ROOT CAUSE	CORRECTIVE ACTION
1 Low distillate rate	• Insufficient heating • Fouled reboiler	• Increase heat input • Clean the reboiler
2 Low product purity	• High boiling impurities in feed • Too fast distillation	• Reduce distillation rate • Improve feed quality
3 High pressure in system	• Condenser blocked • Vent line blocked	• Clean condenser • Clear vent line
4 Temperature fluctuation	• Unstable heat supply • Irregular feed	• Stabilize heat input • Maintain steady feed
5 Foaming in column	• Presence of surface active impurities	• Add antifoam agent • Reduce heat input
6 Bumping in reboiler	• Sudden boiling due to superheating	• Boil gently • Use boiling chips
7 Leakage	• Loose connections / gasket failure	• Tighten connections • Replace gasket
8 Low recovery	• Poor condensation • Heat loss	• Improve cooling • Insulate system

2 FRACTIONAL DISTILLATION

PROBLEM / SYMPTOM	ROOT CAUSE	CORRECTIVE ACTION
1 Low separation efficiency	• Less number of trays • Improper reflux ratio	• Increase number of trays • Optimize reflux ratio
2 Flooding in column	• High vapor velocity • Excessive feed rate	• Reduce vapor rate • Reduce feed rate
3 Weeping	• Low vapor velocity • High reflux ratio	• Increase vapor rate • Reduce reflux
4 Low distillate quality	• Poor reflux • Tray weeping or damage	• Increase reflux ratio • Check and clean trays
5 High pressure drop	• Tray hole blockage • Fouling / scale formation	• Clean trays • Remove scale
6 Irregular level in column	• Unstable feed • Overhead condenser issue	• Stabilize feed • Check condenser
7 Temperature profile irregular	• Faulty temperature sensors • Uneven heat input	• Calibrate sensors • Ensure even heating
8 Low bottom product withdrawal	• Sump level low • Bottom valve partially closed	• Maintain proper level • Fully open bottom valve
9 Entrainment of liquid	• High vapor velocity • Damaged internals	• Reduce vapor rate • Check & repair internals
10 Reboiler fouling	• Scale / polymer deposits • Poor quality feed	• Clean reboiler • Improve feed quality

3 STEAM DISTILLATION

PROBLEM / SYMPTOM	ROOT CAUSE	CORRECTIVE ACTION
1 Low distillate rate	• Insufficient steam supply • Steam leaks	• Increase steam flow • Check and repair leaks
2 Poor oil yield / Low recovery	• Insufficient contact time • High steam rate	• Increase contact time • Optimize steam rate
3 Emulsion formation	• High oil content in layer • Impurities in feed	• Allow proper settling • Use demulsifier
4 Foaming	• Impurities / Surfactants present	• Reduce steam rate • Add antifoam
5 Condensate not separating	• Dirty separator • High oil droplets	• Clean separator • Maintain proper temp.
6 Carryover of water in distillate	• High steam rate • Too much entrainment	• Reduce steam rate • Improve demister
7 Unpleasant odor	• Overheating • Long distillation time	• Reduce heat • Optimize time
8 High pressure in system	• Blocked vent • Condenser issue	• Clear vent line • Check condenser

4 VACUUM DISTILLATION

PROBLEM / SYMPTOM	ROOT CAUSE	CORRECTIVE ACTION
1 High boiling at low temperature	• Vacuum not achieved • Air leaks	• Check vacuum system • Eliminate leaks
2 Low throughput	• Insufficient heating • Fouled reboiler	• Increase heating • Clean reboiler
3 Foaming	• Decomposition of material • Too high vacuum	• Reduce vacuum level • Control temperature
4 Back diffusion of air	• Vacuum breaker failure • Leak in system	• Check vacuum breaker • Repair leaks
5 Bumping	• Sudden pressure change • Superheated liquid	• Boil gently • Use boiling chips
6 Low product quality	• Thermal decomposition • High temperature	• Reduce temperature • Improve vacuum
7 Condenser overload	• High vapor flow • Inadequate cooling	• Increase cooling • Reduce feed rate
8 Oil carryover in distillate	• High vapor velocity • Damaged demister	• Reduce vapor rate • Check demister

5 FLASH DISTILLATION

PROBLEM / SYMPTOM	ROOT CAUSE	CORRECTIVE ACTION
1 Insufficient flash	• Pressure not reduced enough • High initial temperature	• Reduce pressure • Control feed temperature
2 Low vapor yield	• Low feed temperature • High system pressure	• Increase feed temperature • Reduce pressure
3 Carryover of liquid	• High feed rate • Poor inlet device	• Reduce feed rate • Improve inlet device
4 Temperature not as expected	• Incorrect pressure • Temperature control issue	• Check pressure control • Calibrate instruments
5 Fouling / scaling	• Impurities in feed • High solids content	• Pre-treat feed • Clean vessel
6 Unstable operation	• Fluctuating feed • Pressure fluctuations	• Stabilize feed • Improve control
7 Safety valve lifting	• Over pressure • Valve setting low	• Check pressure • Reset valve
8 High energy consumption	• Excessive pressure drop • Heat loss	• Optimize pressure • Insulate system

6 AZEOTROPIC DISTILLATION

PROBLEM / SYMPTOM	ROOT CAUSE	CORRECTIVE ACTION
1 Azeotrope composition not broken	• Improper entrainer • Insufficient entrainer	• Select proper entrainer • Increase entrainer feed
2 Low purity of product	• Low reflux ratio • Insufficient stages	• Increase reflux • Increase stages
3 High entrainer in product	• Poor separation • High reboiler temp.	• Improve separation • Reduce temp.
4 Loss of entrainer	• Carryover in distillate • High boil entrainer	• Improve reflux • Recover entrainer
5 Emulsion formation	• Presence of water • Incompatible liquids	• Remove water • Add demulsifier
6 High energy consumption	• High reflux ratio • Inefficient condenser	• Optimize reflux • Improve condenser
7 Irregular operation	• Unstable feed • Control issues	• Stabilize feed • Check control loops

7 EXTRACTIVE DISTILLATION

PROBLEM / SYMPTOM	ROOT CAUSE	CORRECTIVE ACTION
1 Low separation factor	• Wrong solvent • Less solvent	• Select proper solvent • Increase solvent rate
2 High solvent in product	• Insufficient stages • Low reflux	• Increase stages • Increase reflux
3 Solvent loss	• Carryover in distillate • High vapor rate	• Reduce vapor rate • Improve reflux
4 High energy consumption	• High solvent flow • Inefficient reboiler	• Optimize solvent flow • Improve efficiency
7 Corrosion	• Incompatible solvent	• Select compatible solvent / material
8 Foaming	• Solvent + feed interaction	• Reduce vapor rate • Add antifoam
7 Difficulty in solvent recovery	• Close boiling pts. • Improper column design	• Redesign column • Adjust pressure
8 Poor product purity	• Improper solvent ratio • Insufficient contact	• Optimize solvent/feed ratio • Increase contact time

GENERAL TIPS FOR ALL DISTILLATION SYSTEMS

- Maintain proper feed quality and steady feed rate.
- Ensure proper heat input and uniform distribution.
- Keep condensers and reboilers clean.
- Monitor pressure, temperature and levels regularly.
- Follow preventive maintenance to avoid downtime.

